

Kabadiwala Connect – Bringing the Informal Collector into the Formal Recycling Chain

Background: The overwhelming majority of India's end-of-life electronics is collected through informal scrap dealers, waste-pickers, and local aggregators due to their extensive last-mile reach and low collection costs. However, these informal collectors remain largely outside the formal recycling ecosystem. The E-Waste (Management) Rules, 2022 established a formal Extended Producer Responsibility (EPR) framework involving authorized recyclers, but there is currently limited access for informal collectors to participate in this formal recycling chain. As a result, material collected by informal aggregators may undergo unsafe backyard processing such as open-air cable burning, acid leaching of printed circuit boards, and manual desoldering without proper extraction facilities. While materials such as copper and small quantities of gold may be recovered, valuable materials including lithium, cobalt, neodymium, tantalum, gallium, and indium may be lost. These practices also expose workers to significant health and safety risks. The fundamental gap is not only technological but also informational and institutional. Informal collectors may not know the prevailing fair price of materials, which nearby recyclers are authorized, how to complete a compliant material handover, or how to obtain a documented record of the transaction. Consequently, there is limited incentive for collectors to prefer the formal recycling route.

Problem: Design and develop a vernacular, low-literacy, offline-tolerant mobile platform that enables informal scrap collectors to discover fair prices, connect directly with authorized recyclers, complete a documented and traceable handover of collected materials, and receive payment. The platform should make the formal recycling channel an economically attractive and convenient option for informal collectors rather than creating an additional compliance burden.

The platform should:

- Allow collectors to photograph, categorize, and create digital lots of collected materials such as CRTs, LCD panels, PCBs, cables, batteries, motors and magnet-bearing assemblies, and mixed plastics, enter approximate weight, and receive an instant value estimate.
- Provide a price discovery and historical price dataset containing material category, sub-category, location, date, prevailing buying price, unit of measurement, approximate market range, and recycler/aggregator offered price. The system should use this dataset to provide transparent and current price information to collectors and identify basic price trends.
- Maintain a material and transaction dataset containing lot/reference ID, material category, material description, photograph/image reference, approximate weight, estimated value, quoted price, final sale value, date and time of collection, collection location, recycler details, and transaction status. This dataset should enable traceability of material from collection to authorized recycling.
- Maintain an authorized recycler/aggregator dataset containing recycler/aggregator name, facility location, materials accepted, authorization/registration details, authorization status, contact details, offered rates, pickup availability, and service area. The platform should use this information to identify and rank suitable authorized recyclers for a collector's lot.
- Use the collected material image, category, weight, location, historical price, and transaction data to support AI/ML-based features such as material classification, approximate valuation, recycler

matching, and identification of abnormal or inconsistent transaction values, wherever sufficient training data is available. • Provide current buying rates for different material categories and locations through a simple price board, including spoken price information and basic price trends. • Match collected lots with nearby authorized recyclers or aggregators based on location, material category, offered rate, pickup availability, and authorization status. • Generate a digital and verifiable handover/transfer record containing photographs, weight, timestamp, GPS/location details, and a unique reference that can be confirmed by the recycler. • Maintain an easy-to-understand earnings ledger showing transactions, payments, and pending dues, thereby building a usable financial and transaction history for collectors. • Provide pictorial and/or audio-based safety guidance on hazardous practices, including improper burning or opening of materials and safe handling of batteries and CRTs. • Support Marathi and Hindi at a minimum and provide a genuinely usable interface for users with limited literacy. • Operate in low-connectivity environments through an offline-first architecture, allowing core activities to be completed offline and synchronized when connectivity becomes available. • Support entry-level Android devices with a small application size and low memory requirements. Allow cash-based transactions while keeping digital payment optional and not making it a prerequisite for using the platform.

Dataset Requirements: The solution should be designed to create and utilize structured datasets generated through field operations and platform transactions. The minimum dataset should cover the following:

- **Material Dataset:** Material category, sub-category, material description, image, approximate weight, condition, source type, and estimated value.
- **Price Dataset:** Material category, location, date/time, buying price, selling/quoted price, unit, recycler/aggregator, and historical price information.
- **Recycler Dataset:** Recycler/facility name, location, materials accepted, authorization status/details, contact information, offered rate, pickup availability, and service area.
- **Transaction Dataset:** Unique lot ID, collector ID, material category, quantity/weight, quoted price, final price, recycler ID, collection location, handover location, date/time, payment status, and transaction status.
- **Traceability Dataset:** Lot ID, photographs, weight, timestamp, GPS/location, handover reference number, recycler confirmation, and subsequent transaction status.
- **Collector Dataset:** A minimal profile containing collector ID, preferred language, general operating location, transaction history, and earnings history. The system should avoid collecting unnecessary personal information.
- **AI/ML Training Dataset:** Where AI/ML functionality is proposed, teams should develop or use appropriately sourced datasets containing material images, material categories, weights, prices, locations, and transaction records for model training and validation. Teams should clearly identify the source, quality, size, and limitations of such datasets.

• The dataset should support data cleaning, validation, anonymization where required, historical analysis, price prediction, material classification, recycler recommendation, and transaction-level traceability. • Teams should demonstrate how the dataset is generated, stored, validated, updated, and used by the application rather than treating the dataset as a static database.

Expected Outcome: The proposed solution should create a simple digital bridge between informal collectors and the formal recycling ecosystem, improving price transparency, enabling traceable material handovers, connecting collectors with authorized recyclers, promoting safer handling practices, and encouraging greater participation in the formal recycling chain. The solution

should include a working mobile application, recycler-side interface, structured datasets for materials, prices, recyclers and transactions, field research involving at least two working scrap collectors or aggregators, and a live usability demonstration. Teams should also provide a short unit-economics assessment comparing the collector's existing earnings with the potential earnings through the proposed platform and explaining how the platform can sustain its operations.